An Equilibrium Search Model of Inventory Management

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Introduction

Intermediaries play an important role in many markets.

Facilitate matching between buyers and sellers [search frictions].

Provide liquidity to markets by holding **inventory.**

- Inventory is costly.

- Increases the chance a match can occur when uncertainty exists.

Academic work on intermediaries [retail] typically ignores inventory management.

Yet, inventory management is a big deal. Entire industry (and academic departments) devoted to providing inventory management solutions.

This Paper

1. Present a model of intermediaries that dynamically manage inventory.

a) Buyers and sellers engage in directed search.

b) No ex-ante heterogeneity among buyers, sellers, or dealers.

2. In used car markets, document relationship b/w inventory and,

1. dealer acquisitions and sales;
2. list prices;
3. 1,000s of dealers and weekly info on car listings.

3. Calibrate model.

a) Quantify value of inventory management.
   - One-in one-out policy
   - No dynamic pricing

b) Quantify amount of price dispersion due to inventory management.

Results

1. Model.

- No ex-ante heterogeneity $\rightarrow$ unique pure strategy eqm. with price dispersion.
- Price dispersion: opportunity costs of stochastic demand and inventory management.

2. Empirics – used car dealers.

- Acquisition–inventory elasticity: -0.65.
- Sales–inventory elasticity: 0.85.
- Inventory $\downarrow$ by one car $\implies$ price $\uparrow$ $\$19$.

3. Value of inventory management.

- One-in, one-out policy: 55% of baseline surplus.
- No dynamic pricing: 70% of baseline surplus.
- Price dispersion: Model generates 12–57% of 10-90 inter-pct range of prices.

Contribution to Exisitng Literature

1. **Value of intermediation.**
   - Salz (2020; trash); Gavazza (2016; aircraft); Biglaiser et al (2020; cars)
   - e-commerce: Brown and Goolsbee (2002); Dinerstein et al (2018); more
   - Financial markets: Egan (2019); Robles-Garcia (2019); more

2. **Rationalize price dispersion.**
   Generate price dispersion with ex-ante homogeneous agents.
   - Salop/Stiglitz (1977), Burdett and Judd (1983), Varian (1985).
   - Coey, Larsen, Platt (2019) [Search w/ deadlines]
   - Emp. inv. mgmt: K. Williams (2020); Aryal, Murry, JW. Williams (2021)

3. **Inventory management and search.**
   Connect inventory management with equilibrium search.
   - Directed search: Shi (2009), Menzio and Shi (2011), etc.
   - Revenue mgmt: Aguiregabiria (1999), Hall and Rust (2000)

Model

Model Overview

Buyers purchase indivisible goods from sellers through intermediaries.

Time is continuous and lasts forever.

Buyers: Unit demand, utility $u > 0$, short lived.

Sellers: Unit supply, reservation value 0, short lived.

Intermediaries:

- Buy from sellers in wholesale market, sell to buyers in retail market.
- Hold inventory $x \in \mathbb{N}$ at increasing and convex cost $c(x)$

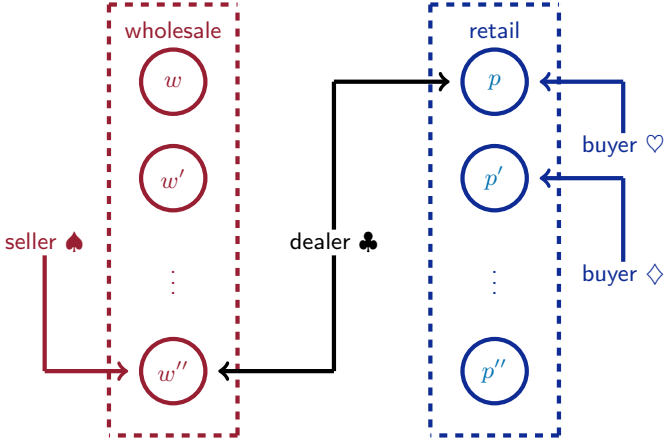
Directed Search and Matching

Retail/wholesale sub-markets:
- indexed by p and w.

Let $\theta(p)$ and $\lambda(w)$ be submarket tightness: $\frac{\#\text{buyers or sellers}}{\#\text{of dealers}}$

CRS matching function.

- meet buyer w.p. $\phi_r(\theta(p))$
- meet seller w.p. $\phi_w(\lambda(p))$



Intermediary/Buyer/Seller Problem

Intermediary: State variable: inventory, x.

$$\rho V(x) = \max_{p,w\geq 0} -c(x) \quad + \quad \underbrace{\phi_r(\theta(p))[p + V(x-1) - V(x)]}_{\text{retail market}} \\ + \quad \underbrace{\phi_w(\lambda(w))[-w + V(x+1) - V(x)]}_{\text{wholesale market}}, \tag{1}$$

Buyers: Free entry in retail sub-market p.

$$\kappa_b = \underbrace{\psi_r(\theta(p))}_{Pr(\text{meet dealer})} (u - p) = \frac{\phi_r(\theta(p))}{\theta(p)}(u - p)$$

Sellers: Free entry in wholesale sub-market w

$$\kappa_s = \underbrace{\psi_w(\lambda(w))}_{Pr(\text{meet dealer})} w = \frac{\phi_w(\lambda(w))}{\lambda(w)} w$$

Equilibrium Characterization I

The equilibrium policy $\theta^*(x), \lambda^*(x)$, given state (inventory) x solve

$$rV(x) = -c(x) + \max_{\theta \geq 0} \Big\{ \underbrace{\phi_r(\theta)[u + V(x-1) - V(x)] - \kappa_b \theta}_{\text{retail transaction surplus}} \Big\} \tag{2}$$

$$+ \max_{\lambda \geq 0} \Big\{ \underbrace{\phi_w(\lambda)[V(x+1) - V(x)] - \kappa_s \lambda}_{\text{wholesale transaction surplus}} \Big\}, \tag{3}$$

and equilibrium prices are given by free-entry conditions

$$p(\theta(x)) = u - \frac{\kappa_b}{\psi_r(\theta(x))} = u - \frac{\kappa_b \theta(x)}{\phi_r(\theta(x))}, \qquad \text{(law of demand } p\text{-}\theta\text{)}$$

and

$$w^*(x) = \frac{\kappa_s}{\psi_w(\lambda^*(x))} = \frac{\kappa_s \lambda(x)}{\phi_w(\lambda(x))}. \qquad \text{(law of supply } w\text{-}\lambda\text{)}$$

Equilibrium Characterization II

Equilibrium is

- retail and wholesale price functions,
- functions describing the retail and wholesale market "tightness",
- intermediary value functions,

that solve the intermediary's problem above.

Block Recursive Equilibrium (Menzio and Shi, 2011)

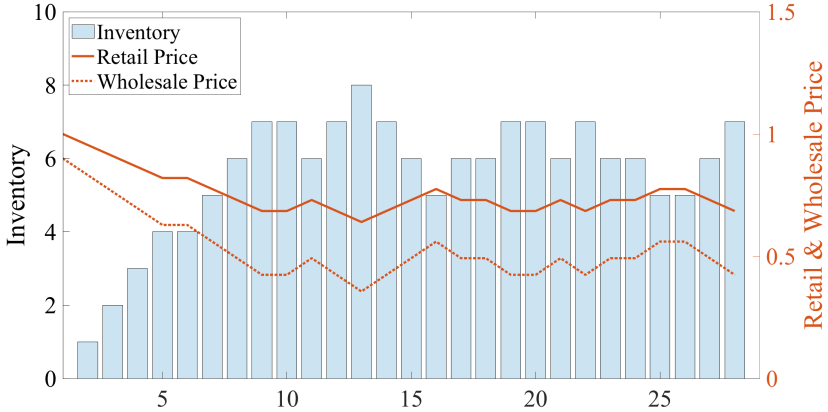
The intermediary's problem does not depend on the distribution of heterogeneous states across all intermediaries.

Ex-post heterogeneity

In Eqm. sellers/buyers "order" themselves by w/p s.t. there is no incentive to change the ordering given steady state inventory.

Model Results: Intermediary Behavior

Single Intermediary



Model Analysis

Inventory Management

- Inventory lowers stock-out risk: $\Rightarrow V(x) \uparrow$ for small x.
- The risk diminishes for large $x \Rightarrow$ benefit $\downarrow \Rightarrow V(x)$ is concave

Price Dynamics

$$\begin{array}{c}\text{Lower } p \text{ to sell quickly.} \\ \text{Lower } w \text{ to order slowly.}\end{array} \rightarrow \text{Retail and wholesale prices co-move.}$$

Search frictions: $\Rightarrow$ sticky adjustment of inventory and price.

Steady State

- Unique, unimodal, steady state distribution of inventory holdings.
- Since $p^*(x)$ is monotone, retail price distribution is unimodal.

Extensions

Product differentiation.

- Extra benefit to hold more inventory (Schevchenko, 2004).

Multi-unit orders.

- Dealer orders according to (s, S) rule.

Used Car Market and Data

Market

Large, active market with very salient inventory management.

Costly search for buyers.

About 2/3 of sales made through dealers.

Data

300,000 cars listed on cars.com in 2018:

- weekly frequency;
- VIN (make, model, year, trim, color, etc);
- mileage, certified pre-owned;
- dealers list entire inventory;
- Focus on "young" non-luxury sedans and SUVs.

Inventory, sales, and acquisitions

Empirical Strategy: Dynamic panel IV.

- Variation: within dealer over time.
- IV: lagged inventory.

Sales / Acquisitions $\leftarrow$ Inventory

$$y_{kft} = \beta x_{kft} + \boldsymbol{Z}_{kft}\boldsymbol{\alpha} + d_{kf} + u_{kft}. \tag{4}$$

where y_{jft} is acquisitions or "sales," k indexes product-categories, f dealers, and t week.

List price $\leftarrow$ Inventory

$$log(p_{kft}) = \beta x_{kft} + \boldsymbol{Z}_{kft}\boldsymbol{\alpha} + d_{kf} + u_{kft}. \tag{5}$$

Descriptive Statistics

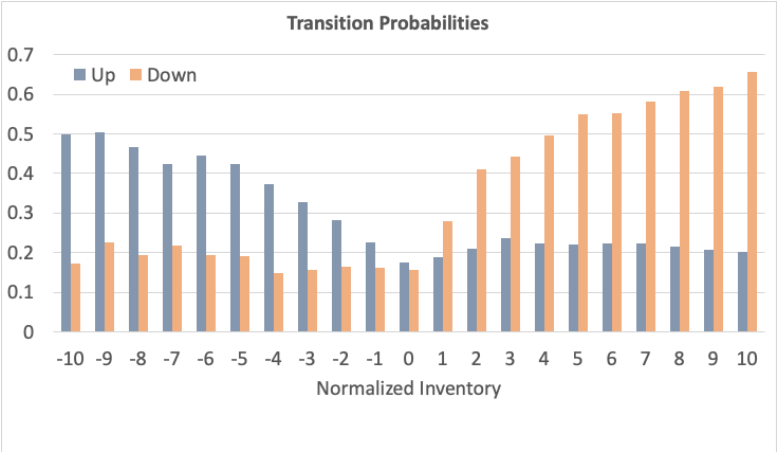
Panel A. Product-dealer level†

	Inventory		New orders		Car removals	
	Average	Volatility	Average	Volatility	Average	Volatility
Mean	13.05	3.76	1.89	1.63	1.89	1.46
Median	8.63	2.80	1.17	1.23	1.19	1.18
SD	15.26	4.78	3.62	2.30	3.60	2.08

Panel B. Car level‡

			Listing prices ($)			
	Listing weeks	Car age	Initial price	Last price	Average	Volatility
Mean	7.83	4.93	16,407	15,702	16,121	376
Median	6	4	15,000	14,500	14,975	251
SD	7.15	3.18	8,040	7,795	7,927	508

Probability of Inventory Transitions



Results

Dealer acquisitions

1% greater inventory at the beginning of a week $\rightarrow$ acquire 0.65% fewer cars.

Sales

1% greater inventory at the beginning of a week $\rightarrow$ "sell" 0.85% more.

List prices

1 additional car at the beginning of week $\rightarrow$ list price drops by \$19.

Large v small dealers

Large dealers twice as elastic in the retail market.

Price elasticity (list) wrt inventory at large dealers 10x larger than small dealers.

▸ tables

What is the Value of Inventory Management?

What is the Value of Inventory Management?

Two sources of value in our model:

1. **Dealers facilitate trading.**

2. **Dealers improve search through active inventory management.**

   (a) Hold multiple units.

   (b) Dynamically price $\rightarrow$ balance supply/demand when search is stochastic.

Social value: avoiding stock-out.

Review of Model and Calibration Details

1. Dealer can only sell and buy 1 item in a *week*. Inventory flow cost $c \cdot x$.
   - Discount at rate ρ.
2. Buyer demand: $p^*(x) = u - \frac{\kappa_b}{\psi_r(\theta^*(x))}$.
3. Seller supply: $w^*(x) = \frac{\kappa_s}{\psi_w(\lambda^*(x))}$.
4. Matching: Urn ball. $\phi_r(\theta) = 1 - e^{\theta}$ and $\phi_w(\lambda) = 1 - e^{\lambda}$

Notes

- Use 4-6 year old sedans.
- Homogeneous product: we residualize prices.
- Homogenous dealers: "Normalize" inventory transitions.
- Consider deviation by a single dealer.

▸ details

Baseline Calibration

Parameter	Value	Description
ρ	$5\%/52$	Weekly interest rate to match a 5% annual rate
u	$16,270$ (\$)	Buyer valuation.
κ_b	$3,732$ (\$)	Buyer's opportunity cost of search.
κ_s	$7,527$ (\$)	Seller's opportunity cost of search.
c	1.282 (\$)	Constant marginal cost of inventory.

	Social surplus	Prob. of trade	Retail price	Wholesale price	Avg. Inventory
Baseline model (BM)	\$ $856,815$	0.29	\$ $11,775$	\$ $8,920$	4.4

The value of holding multiple units.

Counterfactual I: One-in, one-out policy

Assume dealer can hold max of one unit in inventory.

Total surplus decreases to $460,045, or **53.69% of baseline.**

		Social surplus	Prob. of trade	Retail price	Wholesale price
Baseline model		$\$\ 856,815$	0.29	$\$\ 11,775$	$\$\ 8,920$
Fixed inventory capacity $x \in \{0,1\}$	(% Baseline)	53.69%	55.17%	98.75%	98.74%

The value of dynamic inventory management.

Counterfactual II: Shut down dynamic pricing.

Dealers can hold inventory, BUT they can't optimally control inflows and outflows.

		Social surplus	Prob. of trade	Retail price	Wholesale price
Baseline model		$\$\ 856,815$	0.29	$\$\ 11,775$	$\$\ 8,920$
Frozen retail†	(% Baseline)	90.01%	90.46%	–	99.44%
Frozen wholesale‡	(% Baseline)	98.45%	98.95%	100.10%	–
Frozen retail and wholesale w. exits	(% Baseline)	70.05%	–	–	–

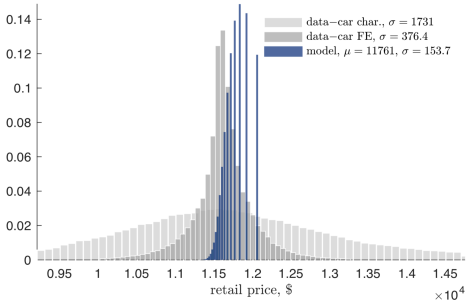
Details: Randomly select dealer and force them to keep current price. Average over density of inventory holdings.

How much price dispersion is due to inventory management?

The 90-10 inter-pct range of prices:

12% of data after residualizing by make/model/age;

57% of the data after residualizing by VIN and time-on-the-market.



Conclusion

Directed search + inventory management $\rightarrow$ price dispersion.

Potentially large welfare implications for inventory management.

Straightforward extensions to more realistic settings.

Caveats / Future Work

- More about the theory of dealer size.
- Heterogeneity across dealers.
- Incorporate other roles of dealers:
  - improved information;
  - market power.

Extra Slides

Calibration Details I

1. Recover eqm. market tightness.

$$M(x, x-1) = \phi_r(\theta(x))$$
$$(x, x+1) = \phi_w(\lambda(x))$$

2. Recover κ's from following relationship.

$$p(\theta(x+1)) = \kappa_s \frac{1}{\phi_w'(\lambda(x))} + \kappa_b \left(\frac{1}{\phi_r'(\theta(x+1))} - \frac{1}{\psi_r(\theta(x+1))} \right).$$

back

Calibration Details II

Now, we need to get c and u.

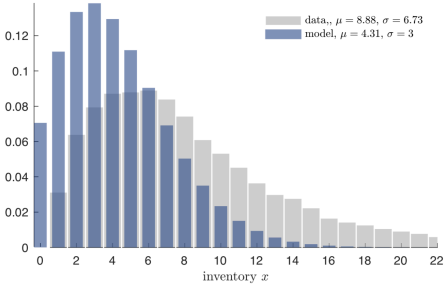
3. Get u from the following relationships.

$$u = p(\theta(x)) + \frac{\kappa_b}{\psi_r(\theta(x))} \quad \text{and} \quad u = \frac{\kappa_s}{\phi'_w(\lambda(x))} + \frac{\kappa_b}{\phi'_r(\theta(x+1))}.$$

4. Then, all that is left is $c \cdot x$. We need to find $V(x) - V(x-1)$.

$$V(x+1) - V(x) = u - \frac{\kappa_b}{\phi'_r(\theta(x+1))} = \frac{\kappa_s}{\phi'_w(\lambda(x))}.$$

back


data,, $\mu = 8.88$, $\sigma = 6.73$
model, $\mu = 4.31$, $\sigma = 3$
0.12
0.1
0.08
0.06
0.04
0.02
0
0
2
4
6
8
10
12
14
16
18
20
22
inventory x

Is 4 small?

```
sum inv_fkt if seg==2, detail

      Percentiles
 5%            2
10%            3        Obs               26,052
25%            4        Sum of Wgt.       26,052

50%            7        Mean            9.19154
                        Std. Dev.      8.369064
75%           11
90%           17        Variance       70.04124
95%           23        Skewness       6.129997
```

Tables I

	New Orders		Listing Removals	
	(1) OLS	(2) IV	(3) OLS	(4) IV
Log Inventory	-0.0518***	-0.6486***	0.4318***	0.8512***
	(0.0043)	(0.0675)	(0.0038)	(0.0597)
... other cars at same dealer	-0.0003***	-0.0029***	-0.0004***	0.0017***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
... rivals in zipcode	0.0041	0.0245***	0.0003	-0.0132***
	(0.0041)	(0.0054)	(0.0037)	(0.0048)
... rivals in all zip-codes	0.0000***	0.0000***	0.0000***	-0.0000***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
R^2				
First Stage F-stat				

back

Dealer Heterogeneity

	(1) Log(Orders)	(2) Log(Removals)	(3) Log(Price)
Log of inventory	-0.5668***	0.5938***	-0.0031
	(0.0712)	(0.0639)	(0.0059)
*Large Dealer	-0.1637	0.5153***	-0.0389***
	(0.1231)	(0.1104)	(0.0066)
Weeks on sale			-0.0107***
			(0.0027)
*Large Dealer			-0.0015***
			(0.0001)
Log of inventory o	-0.0032***	0.0026***	-0.0002***
same dealer	(0.0005)	(0.0004)	(0.0000)
Log of inventory	0.0265***	-0.0196***	0.0013***
same zipcode	(0.0059)	(0.0053)	(0.0002)
Log of inventory	0.0000***	-0.0000***	0.0000***
different zipcodes	(0.00000)	(0.0000)	(0.0000)

back

1st stage: sales and acquisitions

	(1)	(2)	(3)	(4)
Log of inventory two weeks ago	-0.0222***			
	(0.0006)			
Log of inventory four weeks ago		-0.0147***		
		(0.0006)		
Log of inventory six weeks ago			-0.0107***	
			(0.0006)	
Log of inventory two months ago				-0.0067***
				(0.0007)
R square	0.4888	0.2704	0.1671	0.1087

back

1st stage: prices

	(1)	(2)	(3)	(4)
Log of inventory two weeks ago	-0.0220***			
	(0.0001)			
Log of inventory four weeks ago		-0.0191***		
		(0.0001)		
Log of inventory six weeks ago			-0.0161***	
			(0.0001)	
Log of inventory two months ago				-0.0135***
				(0.0001)
R square	0.1394	0.1462	0.1452	0.1418

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